

Presentation
On
**“Transaction & ACID Properties and
their States”**



Presented By

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Transaction

A transaction is a sequence of database operations that run as a single logical unit. All the steps inside a transaction must succeed together. If any step fails, the entire transaction is rolled back, and the database returns to its previous stable state.

Example: Transferring money from one bank account to another involves deducting money from one account and adding it to the other. Both actions must succeed together.

Examples of Transactions

- Transferring money between two bank accounts
- Placing an online shopping order
- Booking a train ticket
- Inserting a new student record while updating related tables

Why Transactions Are Important

- They prevent incomplete or partial updates
- They maintain reliability even if the system crashes
- They make it safe for multiple users to work at the same time
- They keep data consistent and error-free
- They avoid issues such as double deductions or corrupted data

ACID Properties

1. ACID is a set of rules that ensures safe and reliable execution of transactions.
2. ACID stands for Atomicity, Consistency, Isolation, Durability

1. Atomicity (All or Nothing)

Atomicity ensures that all steps in a transaction are treated as a single unit. If any operation fails, the entire transaction is rolled back. No partial updates are allowed.

Example:

If money is deducted from Account A but not added to Account B due to an error, the entire transaction is reversed.

More examples:

- Online payment where deduction and order confirmation both must succeed
- Updating multiple related tables together
- ATM cash withdrawal where the database deduction and cash release must match

2. Consistency (Valid State to Valid State)

Consistency ensures that the database always moves from one valid state to another valid state. All rules, constraints and relationships must be satisfied after the transaction.

Example:

If a bank rule says a balance cannot go negative, no withdrawal transaction should violate this rule.

More examples:

- A student ID in the Marks table must exist in the Student table
- Product quantity cannot drop below zero
- A unique email cannot be inserted twice

3. Isolation (No Interference Between Transactions)

Isolation ensures that multiple transactions happening at the same time do not interfere with each other. Each transaction behaves as if it is running alone.

Example:

If two users try to book the last seat in a train, only one should succeed.

More examples:

- Two ATMs updating the same account balance
- Two users editing product stock simultaneously
- Reading student marks while they are being updated

4. Durability (Permanent Changes)

Durability ensures that once a transaction is committed, its changes are permanent and cannot be lost. The data stays safe even in cases like system crashes or power failures.

Example:

After an online order is confirmed, the record must stay stored even if the server crashes.

More examples:

- Bank balance stays updated after a power cut
- Train ticket booking remains confirmed after restart
- A new student record stays saved permanently

Transaction States

A transaction passes through several states from start to finish:

1. Active State

- The transaction has started.
- Operations like read, write, and calculations are being performed.
- Errors may still occur at this stage.

2. Partially Committed State

- All operations have been executed.
- The transaction is waiting for the final commit.
- If a system failure occurs here, the changes are not saved.

3. Failed State

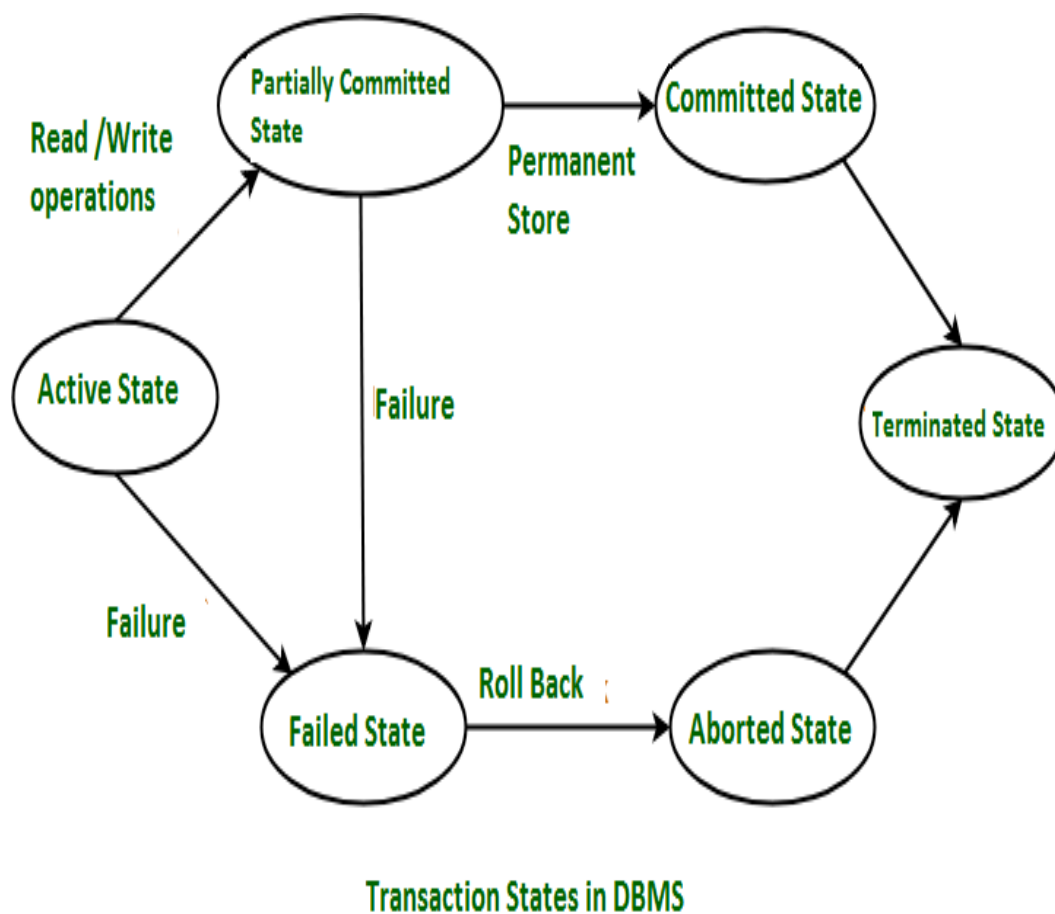
- The transaction cannot proceed due to an error.
- This might be due to logical errors, system failures or constraint violations.

4. Aborted State

- After failure, the transaction is rolled back.
- The database returns to the previous consistent state.
- Optionally, the system may restart the transaction again.

5. Committed State

- The transaction is completed successfully.
- All changes are permanently saved.
- Even if a system crash occurs, the results remain safe.

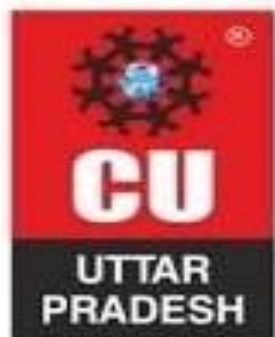


Examples of Transactions

- Bank transfer (debit + credit)
- Online shopping order
- Train ticket booking
- Adding a student record across multiple tables



Assignment File
Data Structure Using C
(25CAH- 604)



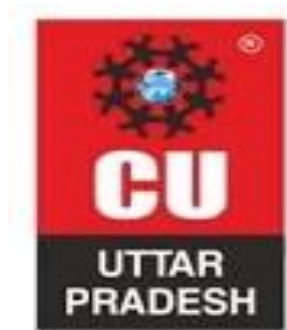
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Assignment File
Artificial Intelligence
(25CAH-601)



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